WEEK 3(b)

```
interface ParkingCharge {
   double calculateCharge();
   void displayParkingDetails();
}

class VehicleParking {
   String vehicleNumber;
   String vehicleType;
   int parkingHours;

   public VehicleParking(String vehicleNumber, String vehicleType, int
parkingHours) {
      this.vehicleNumber = vehicleNumber;
      this.vehicleType = vehicleType;
      this.parkingHours = parkingHours;
   }
}

class TwoWheeler extends VehicleParking implements ParkingCharge {

   public TwoWheeler(String vehicleNumber, int parkingHours) {
      super(vehicleNumber, "Two Wheeler", parkingHours);
   }

  public double calculateCharge() {
```

```
        if (parkingHours <= 2) {
            return parkingHours * 20.0;
        } else {
            return (2 * 20.0) + ((parkingHours - 2) * 10.0);
        }
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println("----------------------------------");
    }
}

class FourWheeler extends VehicleParking implements ParkingCharge {

    public FourWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Four Wheeler", parkingHours);
    }

    public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 50.0;
        } else {
```

```
            return (2 * 50.0) + ((parkingHours - 2) * 30.0);
        }
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println("----------------------------------");
    }
}

public class Main{
    public static void main(String[] args) {

        ParkingCharge parkingRef;

        parkingRef = new TwoWheeler("KA-01-AB-1234", 5);
        parkingRef.displayParkingDetails();

        parkingRef = new FourWheeler("KA-05-CD-5678", 4);
        parkingRef.displayParkingDetails();

    }
}
```

OUTPUT:

```
Vehicle Number: KA-01-AB-1234
Vehicle Type: Two Wheeler
Parking Hours: 5 hrs
Total Parking Charge: Rs. 70.0
-----------------------------------
Vehicle Number: KA-05-CD-5678
Vehicle Type: Four Wheeler
Parking Hours: 4 hrs
Total Parking Charge: Rs. 160.0
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```